

- 1.530 Shane thinks of three integers. Added two at a time, their sums are 37, 41, and 44. What is the product of the three integers?
- 1.530 We are given enough information to write three equations: $a + b = 37$, $b + c = 41$, and $a + c = 44$. Subtract the first from the second to get $c - a = 4$. Add this to $a + c = 44$ and we get $2c = 48$, so $c = 24$. Plug-in to get $a = 20$ and $b = 17$. $abc = 8,160$.
- 1.340 Keith wants to mark all of the original prices in his store down to half-off. The prices are already marked down by 20%. What additional percent should be discounted? Express your answer as a percent rounded to the tenth.
- 1.340 The prices are currently 80% of their original value, so we look for a value x where $0.8x = 0.5$. This gives us $x = 5/8 = 0.625$ or 62.5% of the already marked-down prices, which is an additional **37.5%** off.

- 1.440 On a long hike, Greg forgets to fill his canteen. Fortunately, Harold has 40 ounces of water and Jake has 60 ounces. The three hikers share the water so that each gets an equal amount. At the end of the hike, Greg gives Harold and Jake \$5 to split for the water they gave him. How much should Harold get?
- 1.440 Between them they have 100 ounces, so each gets $100/3$ ounces. Harold has $120/3$ ounces, so he gives Greg $20/3$ ounces, while Jake has $180/3$ ounces and must give Greg $80/3$ ounces. Harold contributed $1/5$ of the water that Greg drank, so he should get \$1 (which is $1/5$ of the money Greg gave them).
- 1.640 Rebecca runs a quarter-mile each week. In each of the last 6 weeks her time has improved (decreased) by 2%, and she can now run a quarter-mile in 66 seconds. What was her time six weeks ago? Round your decimal answer to the tenth of a second.
- 1.640 If her time decreases by 2% six times, her new time will be $(0.98)^6$ times what her original time was. This means that if x is her original time, then

$$x(0.98)^6 = 66, \text{ so } x = \frac{66}{(0.98)^6} \approx 74.5 \text{ seconds.}$$

1.550 Franklin rides his bicycle to the beach averaging 18mph. He rides home along the same route. What speed must he average on the way home if he wants to average 21 miles per hour for the round trip?

1.550 Let's put the formula for harmonic mean to work here, using r to represent the rate in mph that Franklin must ride on the return trip:

$$\frac{2 \cdot 18 \cdot r}{18 + r} = 21, \text{ so } 36r = 378 + 21r.$$

Solving for r we get **25.2 mph**.

1.650 Congruent circles of radius 6 have their centers at $C(5, 4)$ and $D(1, 2)$. The circles intersect at points A and B. What is the slope-intercept form equation of the line which passes through A and B?

1.650 The line which passes through the two points of intersection is the perpendicular bisector of the segment which connects the centers of the circles (this is easiest to see by connecting all four points to make a rhombus, in which the diagonals are perpendicular bisectors, see p.183). The slope between the centers of the circles is $1/2$, so the perpendicular slope is -2 . The midpoint is $(3,3)$. (Average the x and y coordinates to get the midpoint.) The equation of the line which passes through $(3,3)$ with a slope of -2 is $y = -2x + 9$.

- 1.121 The perimeter of a rectangle is 16cm, and the area of the same rectangle is 8cm^2 . What is the diagonal length of the rectangle? (See section 5.3).
- 1.121 If we label the sides of the rectangle a and b , we have perimeter $2(a + b) = 16$ so $a + b = 8$, and the area of the rectangle is $ab = 8$. By the Pythagorean theorem, the diagonal length we are looking for will be $\sqrt{a^2 + b^2}$. We do not have to solve for a and b separately once we recognize that:
 $a^2 + b^2 = (a + b)^2 - 2ab = (8)^2 - 2(8) = 48$, which makes the diagonal length $\sqrt{48} = 4\sqrt{3}$.
- 1.741 The surfaces of a rectangular prism have areas of 12cm^2 , 16cm^2 , and 27cm^2 . What is the volume of the prism?
- 1.741 Call the edge lengths of the prism a , b , and c . We are given $ab = 12$, $ac = 16$, and $bc = 27$ and asked to find abc . If we multiply all three of our equations together we get $a^2b^2c^2 = 12 \cdot 16 \cdot 27$. To get abc , we take the square root of both sides of the equation.
 $abc = \sqrt{12 \cdot 16 \cdot 27} = \sqrt{4 \cdot 16 \cdot 81} = 2 \cdot 4 \cdot 9 = 72\text{cm}^3$.

1.251 The mean of a set of seven positive integers is 7, the median is 8, and the distinct mode is 9. What is the greatest possible range for the set of integers?

1.251 Start with 7 integers labeled in order a, b, c, d, e, f , and g . We are given the median: $d = 8$. The distinct mode of 9 means that e and f must be 9 (or e, f , and g could all be 9). The mean gives us the sum: 49. To maximize the range we want a, b , and c to be as small as possible, allowing f to be as large as possible while keeping the sum equal to 49. Make a, b , and c equal to 1, 2, and 3. This makes the set (whose sum is 49) 1, 2, 3, 8, 9, 9, and f .
 $1 + 2 + 3 + 8 + 9 + 9 + f = 49$, so $f = 17$ and the range is $17 - 1 = 16$.

1.351 Four integers are added to the set $\{3, 4, 5, 5, 8\}$, increasing the mean, median, and mode each by 1. What is the greatest integer in the new set?

1.351 The current set $\{3, 4, 5, 5, 8\}$ has a mean, median, and mode of 5. Of the four integers added to the set, at least three must be 6's to make the mode 6. The mean of the new set must be 6 as well, which means the sum of the integers in the set must be 54. The 7 integers we know already have a sum of $3 + 4 + 5 + 5 + 6 + 6 + 6 + 8 = 43$. This means that the fourth (and largest) number added to the set must be $54 - 43 = 11$. The new set: $\{3, 4, 5, 5, 6, 6, 6, 8, 11\}$.

1.551 In a set of ten integers, the average of the two smallest integers is 102, the average of the three smallest integers is 103, the four smallest is 104, and so forth until the average of all ten integers is 110. What is the greatest of these integers?

1.551 This is a case where guess-and-check can save some time if we recognize that the integers can be 101, 103, 105, 107... until the 10th integer is 119. A more rigorous approach probably looks something like this: Call the integers $a, b, c, d, \dots, i, j$.

$$a + b = 2(102) = 204$$

$$a + b + c = 3(103) = 309, \text{ so } c = 105.$$

$$a + b + c + d = 4(104) = 416, \text{ so } d = 107.$$

... until

$$a + b + c + d + \dots + i = 9(109) = 981, \text{ and}$$

$$a + b + c + d + \dots + j = 10(110) = 1,100, \text{ so } j = \mathbf{119}.$$

1.161 The 5th term of an arithmetic sequence is 18 and the 9th term is 24. What is the 99th term?

1.161 The common difference is found by $\frac{24-18}{9-5} = \frac{6}{4} = \frac{3}{2}$.

We must add this common difference 90 times to get

$$\text{from the 9th to the 99th term: } 24 + 90 \cdot \frac{3}{2} = \mathbf{159}.$$

1.661 Find the perimeter of the smallest triangle whose integer side lengths form a geometric sequence whose common ratio r is not equal to 1.
(See p.178).

1.661 After some quick guessing-and-checking, we see that the common ratio must certainly be less than 2 (otherwise, the long side is always greater than the sum of the two shorter sides, as in 1-2-4). Using a common ratio of $3/2$ allows us to use 4-6-9 as the lengths of the sides, in which case the perimeter of the triangle is $4 + 6 + 9 = 19$ units.

1.171 Simplify:
$$\frac{5}{6 + \frac{5}{6 + \frac{5}{6 + \dots}}}$$

1.171 For $x = \frac{5}{6 + \frac{5}{6 + \frac{5}{6 + \dots}}}$ we have $x = \frac{5}{6 + x}$ or

$x(6 + x) = 5$. Solve the quadratic $x^2 + 6x - 5 = 0$ to get $x = -3 \pm \sqrt{14}$. Only the positive solution makes sense: $x = \sqrt{14} - 3$.

1.271 Simplify: $20 + \sqrt{20 + \sqrt{20 + \sqrt{20 + \dots}}}$.

1.271 Forget the first 20 momentarily and just let

$x = \sqrt{20 + \sqrt{20 + \sqrt{20 \dots}}}$, then $x^2 = 20 + x$, so
 $x^2 - x - 20 = 0$. This factors into $(x - 5)(x + 4) = 0$,
so $x = 5$ (it is assumed that we are looking for the
positive square root). Add the 20 from the beginning
of the expression and we get **25**.

1.471 The formula for the sum of the powers of 2 from 2^0 to 2^n is $2^0 + 2^1 + 2^2 \dots + 2^n = 2^{n+1} - 1$. Write a formula which could be used to find the sum of the powers of a^0 through a^n in terms of a and n .

1.471 Consider the sum $a^0 + a^1 + a^2 + a^3 \dots + a^n$:

if $x = a^0 + a^1 + a^2 + a^3 \dots + a^n$,

then $ax = a^1 + a^2 + a^3 + a^4 \dots + a^{n+1}$, and

$ax - x = a^{n+1} - a^0$. Factor out the x on the left:

$x(a - 1) = a^{n+1} - 1$ and divide by $(a - 1)$ to get:

$$x = \frac{a^{n+1} - 1}{a - 1}.$$

1.571 Simplify: $1 + \frac{3}{2 + \frac{3}{1 + \frac{3}{2 + \frac{3}{1 + \dots}}}}$.

1.571 Let $x = 1 + \frac{3}{2 + \frac{3}{1 + \frac{3}{2 + \frac{3}{1 + \dots}}}}$ Then: $x = 1 + \frac{3}{2 + \frac{3}{x}}$

$$= 1 + \frac{3}{\frac{2x+3}{x}} = 1 + \frac{3x}{2x+3}, \text{ so } x-1 = \frac{3x}{2x+3}, \text{ and}$$

$2x^2 + x - 3 = 3x$. Subtract $3x$ to get the quadratic

$$2x^2 - 2x - 3 = 0, \text{ which has solutions: } x = \frac{1 \pm \sqrt{7}}{2}.$$

Only the positive solution makes sense, so:

$$x = \frac{1 + \sqrt{7}}{2}.$$

- 1.081 The product of two numbers is 30 and their sum is 16.
Express their difference in simplest radical form.

1.081 We have the equations: $a + b = 16$ and $ab = 30$.

Solve the first equation for a : $a = 16 - b$ and substitute this into our second equation to get

$(16 - b)b = 30$ which gives us the quadratic:

$b^2 - 16b + 30 = 0$. It is not factorable, so we substitute values into the quadratic formula and solve:

$$x = \frac{16 \pm \sqrt{256 - 4(30)}}{2} = \frac{16 \pm \sqrt{136}}{2} = \frac{16 \pm 2\sqrt{34}}{2} = 8 \pm \sqrt{34},$$

giving us two roots whose difference is

$$(8 + \sqrt{34}) - (8 - \sqrt{34}) = 2\sqrt{34}.$$

- 1.581 For positive numbers a and b , the positive difference of a and b is divided by the positive difference of their reciprocals and the result is 5. What is the product of a and b ?

1.581 To begin, write an equation to model the question:

$$\frac{\frac{a-b}{1} - \frac{1}{1}}{\frac{1}{b} - \frac{1}{a}} = 5 \text{ means that } \frac{a-b}{\left(\frac{a-b}{ab}\right)} = 5, \text{ so}$$

$$a - b \cdot \frac{ab}{a - b} = 5 \text{ giving us } ab = 5.$$

1.781 Tejas begins to paint a fence that is 100 meters long at noon. By 2pm, Tejas has one third of the fence painted and Jason arrives to help. At 3pm, four-fifths of the fence is painted and Tejas leaves Jason to finish painting the remaining fence. What time does Jason finish painting the fence?

1.781 Take this one step at a time. Tejas paints $\frac{1}{3}$ of the fence in 2 hours, so he paints at a rate of $\frac{1}{6}$ fence per hour. Working together, Tejas and Jason finish $\frac{4}{5} - \frac{1}{3} = \frac{7}{15}$ of the fence, of which $\frac{1}{6}$ was completed by Tejas, meaning that on his own, Jason can complete $\frac{7}{15} - \frac{1}{6} = \frac{3}{10}$ on his own in one hour. Only $\frac{1}{5}$ of the fence is left to be painted, which will take Jason $\frac{1/5}{3/10} = \frac{2}{3}$ of an hour or 40 minutes. He will finish painting at **3:40pm**.

- 1.391 A solid rubber ball rebounds to three-fourths the height it is dropped from. What is the total vertical distance traveled by a ball dropped from a height of 24 feet as it bounces to a rest (for example, when it hits the floor for the second time it will have traveled $24 + 18 + 18 = 60$ feet).

- 1.391 We are looking for the sum of the terms:

$24 + 18 + 18 + 13.5 + 13.5 + \dots$ which resembles an infinite geometric series (except that all but the first term is repeated). Remove that first term (for now) and solve for $36 + 27 + 20.25 \dots$ which is an infinite geometric series with common ratio $3/4$ and first term 36. Use the formula (p. 56) to get the total

distance (less the initial drop): $d = \frac{36}{1 - 0.75} = 144$ ft,

then add back the 24-foot initial drop to get **168** feet.

- 1.491 If $ab = 4$, $bc = 5$, and $ac = 10$, find $a^2 + b^2 + c^2$.

- 1.491 We can find a^2 , b^2 , and c^2 separately using the following:

$$a^2 = \frac{ab \cdot ac}{bc} = \frac{40}{5} = 8$$

$$b^2 = \frac{ab \cdot bc}{ac} = \frac{20}{10} = 2$$

$$c^2 = \frac{bc \cdot ac}{ab} = \frac{50}{4} = 12.5 \text{ so } a^2 + b^2 + c^2 = \mathbf{22.5}$$

1.921 For positive integers a and b , $(a + \sqrt{b})^2 = 9 + 4\sqrt{5}$,
what is $(a + \sqrt{b})^3$? (See section 1.7).

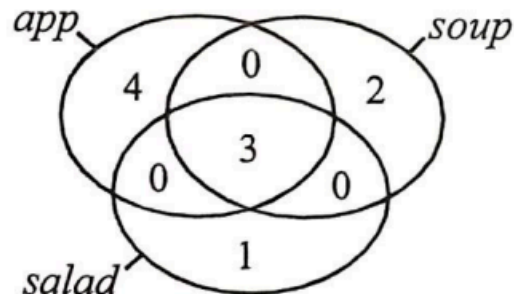
1.921 $(a + \sqrt{b})^2 = a^2 + 2a\sqrt{b} + b = 9 + 4\sqrt{5}$, so b must be 5
(the value under the radical) and a must be 2 (be-
cause $2a\sqrt{5} = 4\sqrt{5}$). To find $(a + \sqrt{b})^3$ we just
multiply $(2 + \sqrt{5})$ by $(9 + 4\sqrt{5})$:
 $(2 + \sqrt{5})(9 + 4\sqrt{5}) = 18 + 8\sqrt{5} + 9\sqrt{5} + 20$
 $= 38 + 17\sqrt{5}$.

2.210 How many distinct pairs of consecutive integers have a product less than 10,000?

2.210 The problem asks for integers, and consecutive negative integers have a positive product so we cannot forget to count them. We can start with $-100(-99)$ and get all the way up to $99 \cdot 100$. The first integer in each pair can be anywhere from -100 to 99 (inclusive). $99 - (-100) + 1 = 200$ pairs of consecutive integers whose product is less than 10,000.

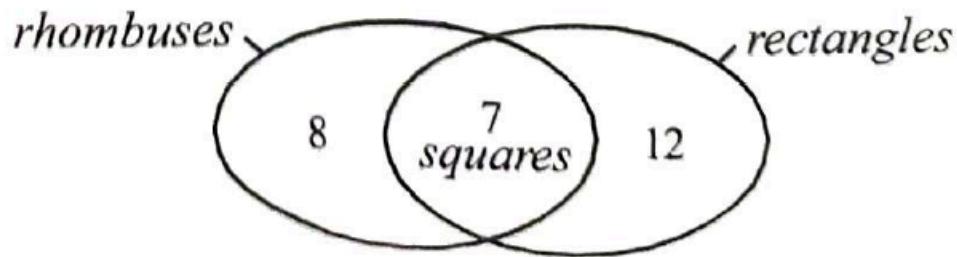
2.120 Ten friends go out to dinner together: 7 order an appetizer, 5 order a soup, and 4 order a salad. If everyone orders something, but no one orders exactly two things, how many people order all three things?

2.120 We can solve this problem with or without a Venn diagram. The number in the overlapping region will be counted three times (two extra times). When we add the number of appetizers, soups, and salads, we get 16. This is 6 more than the number of people. Because no person ordered just two items, 3 items must have been counted two extra times.



2.320 All squares are both rectangles and rhombuses. All rhombuses and rectangles are parallelograms. On a sheet of paper Josh draws 19 rectangles, 15 rhombuses, and 7 squares. How many parallelograms did Josh draw?

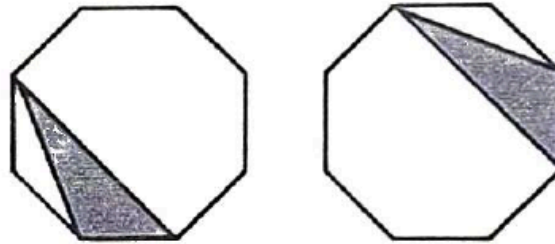
2.320 A Venn diagram helps us organize. The area in the diagram where rectangles and rhombuses overlap represents squares. Fill this region in first then complete the diagram. All of these shapes are parallelograms. There are **27** parallelograms.



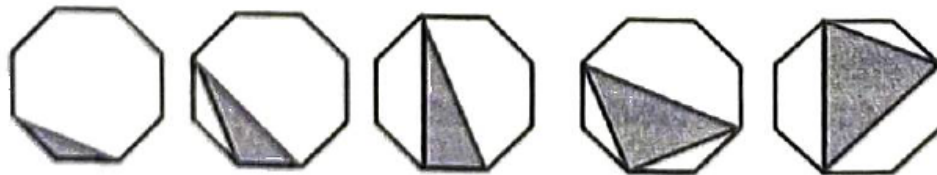
2.520 Students were surveyed on their preference for tea. Three-fourths of the students surveyed like sweet tea, three-fifths of the students surveyed like unsweetened tea, and one sixth of the students do not like either. What is the fewest number of students surveyed who could like both sweet and unsweetened tea?

2.520 Using a common denominator, we have $\frac{45}{60}$ students who like sweet tea, $\frac{36}{60}$ like unsweet tea, and $\frac{10}{60}$ like neither. This gives us a total of $\frac{91}{60}$, which means that $\frac{31}{60}$ of students must like both sweet and unsweet tea. The fewest number of students who could have been surveyed is 60, so at least 31 students must like both.

- 2.640 How many non-congruent triangles can be formed by connecting three of the vertices of a regular octagon. For example, the two triangles drawn below are congruent and should only be counted once.



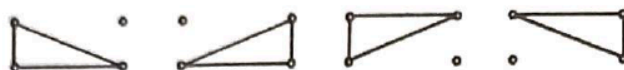
- 2.640 Somewhat surprisingly, there are only 5:



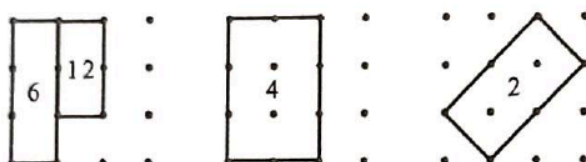
2.840 How many right triangles can be formed by connecting three points on the unit grid shown?



2.840 First, note that for any four points which form a rectangle, there 4 right triangles:

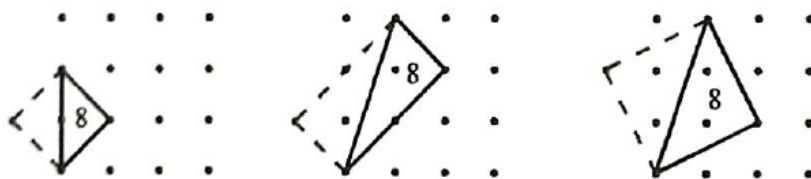


Now we just need to count rectangles. We have already counted the squares (there are 20 in the example on the same page). Now we look for rectangles that are not squares:



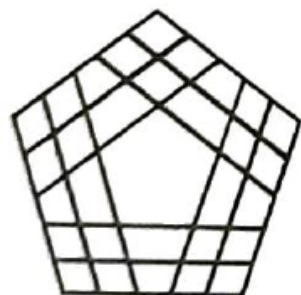
This gives us 44 rectangles (including squares) with four right triangles each, for a total of 176.

Unfortunately, we missed some triangles that are not part of any of these rectangles (in fact, I missed all of the triangles below in the first printing of this book, there are 8 of each type).



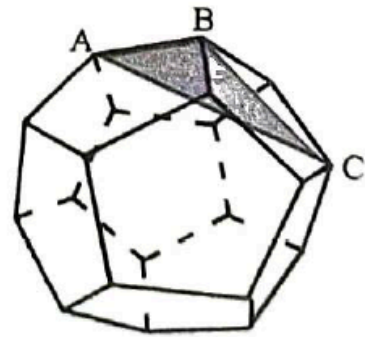
This brings our total to **200**.

2.811 How many pentagons can be formed by tracing the lines of the figure below?



2.811 Every possible pentagon uses one of the three parallel lines on each side. This gives us three choices for each of 5 sides. $3^5 = 243$ pentagons.

- 2.911 A regular dodecahedron has 12 pentagonal faces, 20 vertices, and 30 edges. How many of the triangles which can be formed by connecting three of its vertices have at least one side within the dodecahedron?



- 2.911 With 20 vertices, there are $20C3 = 1,140$ ways to choose three points as vertices of a triangle. There are $5C3 = 10$ of these triangles on each of the 12 pentagonal faces, for a total of 120 triangles on the faces of the dodecahedron. Then, from any vertex, you can go in three different directions along the edges to another vertex. We can connect these three vertices to form an equilateral triangle where each side is a face diagonal. There are 20 of these (one per vertex). So, there are a total of $1,140 - 120 - 20 = 1,000$ triangles.

Example: Corey and Tony are friends on the same basketball team. There are eight players on the team. How many starting lineups of five players include Corey, Tony, or both?

Reasoning: Again, it requires a bit of computation to figure out how many teams include just Corey, just Tony, or both Corey and Tony. Instead, begin by considering how many starting lineups are possible without restrictions (choosing 5 from a team of 8: $8C5 = 56$ ways). Subtract lineups that do not include Corey or Tony (choosing a team of 5 from the remaining 6 players: $6C5 = 6$ ways). Better still, we can choose one of the 6 remaining players to join Corey and Tony on the bench in $6C1 = 6$ ways.

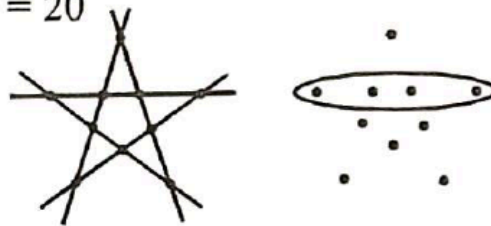
$$\binom{8}{5} - \binom{6}{1} = 56 - 6 = 50 \text{ starting lineups.}$$

We are really trying to count what we don't want, which is often easier to find and subtract from the total.

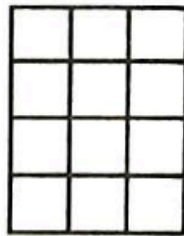
- 2.121 There are seven parking spaces in a row which must be assigned to four co-workers. How many ways can the spaces be assigned if at least two of the assigned spaces must be adjacent?
- 2.121 There are ${}^7C_4 = 35$ ways to choose four spaces, and only one of these arrangements does not have any spaces next to each other (leaving 34 combinations). Each of these arrangements has $4!$ ways to assign coworkers to the spaces, so we have $34(4!) = 816$ ways to assign the four spaces.

2.621 Five lines are drawn so that each intersects the other four, but no three lines intersect at the same point. Each of the points of intersection is then connected to every other point of intersection. How many triangles are formed which have three of the points of intersection as vertices?

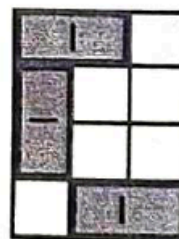
2.621 When we draw five lines with each intersecting the other four, we get $(5 \cdot 4)/2 = 10$ points of intersection (this is a handshake problem). The simplest way to do this is to draw a star. Connecting three of these points to create a triangle requires that we select 3 of the 10 points of intersection $10C3 = 120$ ways, however, there are 5 sets of 4 collinear points (selecting three of these will not make a triangle). We must subtract $5(4C3) = 20$ sets of points which will not form a triangle, leaving us with **100** triangles.



- 2.441 Three identical dominoes are placed on the 3×4 tiled board below to create a complete pathway from the upper-left corner to the lower-right corner of the board (no dominoes may touch only at the corners). How many ways can this be done?



Yes



No

- 2.441 Every 6-unit path from the top left corner to the bottom right corner can be covered with three dominoes. Starting from the top we must find a path which goes right twice and down three times. There are $5C_2 = 10$ ways to arrange 2 moves right among 3 down.

- 2.461 Three friends: Al, Bill, and Carl, are painting their chests with the word CAVALIERS for a basketball game, with at least two letters painted on each friend's chest. For example, Al might get "CA", Bill "VAL", and Carl "IERS", or Carl could get "CAVA" while Bill gets "LI" and Al gets "ERS". How many ways can the three friends get their chests painted so that when they stand in the correct order, they spell out the word CAVALIERS?
- 2.461 The first man must have at least CA on his chest, and the last man must have at least RS. This leaves VALIE for the middle man. You can find all 10 ways to assign VALIE with careful casework or consider placing 2 dividers, each in one of these 6 places: _V_A_L_I_E_. There are $6C_2 = 15$ ways, but we cannot have |V|ALIE or V|A|LIE or any of the 5 arrangements which leave just one letter for the middle man. This gives us 10 ways to divide the letters and there are $3! = 6$ ways to choose who stands 1st, 2nd, and 3rd for a total of $10 \cdot 6 = 60$ ways.

2.561 An elaborate code consists of left and right arrows which are placed together to represent words. There are spaces between words. For example, the code of three left and four right arrows arranged with two spaces: $\ll \gg \diamond$ represents a three-word code phrase. How many different three-word code phrases in this system use 8 letters?

2.561 There are $2^8 = 256$ ways to assign 8 arrows either left or right. We must then place gaps between arrows to separate them into words. Consider the set of 8 arrows: $\gggggggg$. There are $7C2 = 21$ ways to choose two spaces in this (or any other) arrangement, giving us $256 \cdot 21 = \mathbf{5,376}$ code phrases.

2.271 Find $a + b$ if $\binom{9}{5} + \binom{9}{6} = \binom{a}{b}$.

2.271 $\binom{9}{5}$ and $\binom{9}{6}$ are entries on the 9th row of Pascal's triangle. The sum of these two entries is the entry on row 10 which falls between the two: $\binom{10}{6}$.

$10 + 6 = \mathbf{16}$. The rule: $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$.

2.081 How many ways can six slices of pepperoni pizza and three slices of cheese pizza be divided among five students if one of the students is a vegetarian?

2.081 The 6 slices can be divided among the 4 students who are not vegetarians. Use 3 dividers and 6 slices which gives us $9C3 = 84$ ways to divide the pepperoni slices. The 3 slices of cheese pizza can be divided among all 5 students. Using 4 dividers and 3 slices we get $7C3 = 35$ ways to distribute the cheese slices. This gives us a total of $35 \cdot 84 = 2,940$ ways to distribute all 9 slices.

2.181 How many integers between 1,000,000 and 2,000,000 have at least four fives?

2.181 After the 1 in 1,000,000 there are 6 digits. At first it seems that we can simply place 4 fives $6C4 = 15$ ways and then choose any one of 10 digits for each of the remaining two places $10^2 = 100$ ways for a total of 1,500 ways. However, we over-count when there are more than 4 fives. We must count three cases. When there are exactly four 5's there are $6C4 = 15$ times $9^2 = 81$ ways to fill the remaining two places (with something other than a 5) = 1,215 numbers. With exactly five 5's there are $6C5 = 6$ ways to place the fives and 9 ways to fill the remaining place = 54 numbers. There is only one way to have six 5's. This gives us a total of $1,215 + 54 + 1 = 1,270$ numbers.

2.381 Three standard six-sided dice are rolled: one red, one green, and one white. How many outcomes are possible where the number rolled on the red die is greater than the number rolled on the green die, and the number on the green die is greater than the number rolled on the white die?

2.381 No two of the numbers rolled can be the same. There are $6C3 = 20$ ways to select three numbers (from the six possible) and there is one way to order each of these from least to greatest for white/green/red. This leaves us with just **20** ways.

3.500 What is the probability that the top four cards of a shuffled deck are all of the same suit?

3.050 It is easier to count the number of ways to roll a 4 or less. There is one way (1,1) to roll a 2, there are 2 ways to roll a 3 (1,2) and (2,1), and 3 ways to roll a 4 (1,3), (2,2), and (3,1) for a total of 6 out of the $6^2 = 36$ possible rolls, leaving 30 ways to roll greater than a 4. $30/36 = 5/6$.

3.700 What is the probability that the top card in a shuffled deck is an Ace (of any color), and the second card is a spade? The Ace may be the Ace of spades.

3.700 There are two cases to consider: the first card can be an ace of spades, or an ace from another suit. If the ace is the ace of spades, the probability is:

$$\frac{1}{52} \cdot \frac{12}{51} = \frac{12}{52 \cdot 51}.$$

If the top card is the ace of another suit we have:

$$\frac{3}{52} \cdot \frac{13}{51} = \frac{39}{52 \cdot 51}.$$

Added together (now you see why I didn't multiply the $52 \cdot 51$):

$$\frac{51}{52 \cdot 51} = \frac{1}{52}.$$

3.510 A dot is added to each face of two standard dice. What is the probability of rolling an 8 with the modified pair of dice?

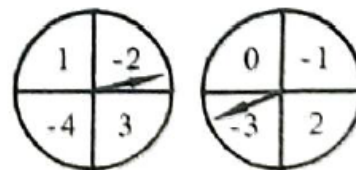
3.510 The modified die will have faces with 2, 3, 4, 5, 6, and 7 dots. There will still be 36 possible rolls with a pair of the modified dice. Of those 36 rolls, 5 will show an 8: (2,6), (3,5), (4,4), (5,3), (6,2), making the probability of rolling an 8: $5/36$.

3.610 For three randomly selected positive integers, what is the probability that the sum of the units digits is even?

3.610 The units digit of each integer can be odd or even, with equal probability. For the sum of the three units digits to be even, we can have either 2 odd digits and 1 even digit: {O,O,E}, {O,E,O}, or {E,O,O}, or we can have 3 even digits: {E,E,E}. There are $2^3 = 8$ possibilities (think of this as counting coin flips, where each flip determines whether the units digit is odd or even), which makes the probability that the sum is even $4/8 = 1/2$.

3.810 Each spinner below is divided into four equal sectors.

Each is spun once, and the product of the spins is calculated. What is the probability that the product is positive?



3.810 The product will be positive if the two spins are either both positive or both negative.

$$P(\text{both positive}) = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}.$$

$$P(\text{both negative}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Add these together to get a probability of $\frac{3}{8}$.

Example: Lisa shakes her piggy bank which contains 5 quarters and 5 nickels until 5 coins fall out. Assuming nickels and quarters fall out with equal probability, what is the probability that the sum of the change that has fallen out is \$0.85?

Reasoning: The only way for five Nickels (N) and Quarters (Q) to add up to \$0.85 is with three Quarters and two Nickels. Begin by computing $P(QQQNN)$:

$$\frac{5}{10} \cdot \frac{4}{9} \cdot \frac{3}{8} \cdot \frac{5}{7} \cdot \frac{4}{6} = \frac{5}{126}.$$

However, there are $5C2 = 10$ ways to arrange $QQQNN$, each has an equal probability of occurring:

$$\frac{5}{126} \cdot 10 = \frac{50}{126} = \frac{25}{63}.$$

Example: There are a dozen eggs in a basket and four have been hard-boiled. You can tell by spinning an egg whether it has been hard-boiled. You select three eggs from the basket. What is the probability that only one of the three is hard-boiled?

Reasoning: The probability of selecting a hard-boiled egg followed by two raw eggs is given by:

$$P(HRR) = \frac{4}{12} \cdot \frac{8}{11} \cdot \frac{7}{10} = \frac{28}{165}.$$

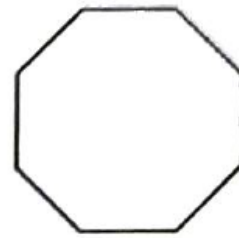
There are two other equally likely possibilities: $P(RHR)$ and $P(RRH)$, so we triple the probability above to get:

$$3 \cdot \frac{28}{165} = \frac{28}{55}.$$

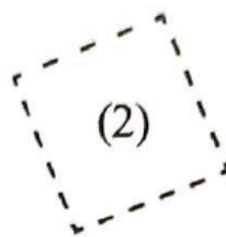
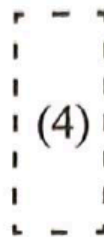
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- 3.030 Three numbers are selected at random and without replacement from a bag containing each of the numbers 1 through 9. What is the probability that the three numbers are consecutive integers?
- 3.030 There are $9C3 = 84$ ways to choose 3 digits from a set of 9. Of these 84 ways, there are 7 sets of consecutive integers (starting with 1-2-3 and ending with 7-8-9). $7/84 = 1/12$.

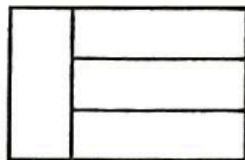
- 3.130 Four of the vertices of a regular octagon are selected at random and connected to form a quadrilateral. What is the probability that the quadrilateral will be a rectangle?



- 3.130 There are $8C4 = 70$ ways to select 4 of the 8 vertices. Only 6 of these combinations form rectangles (shown), making the probability $6/70$ or $3/35$.



- 3.230 An organization is choosing colors for the stripes on its flag. The flag has three horizontal stripes and a vertical stripe as shown. If the color of each of the four stripes is selected at random from the colors red, blue, and green. What is the probability that no two stripes of the same color share an edge?



- 3.230 The total number of ways colors can be chosen for the flag is $3^4 = 81$ (four stripes can each be one of three colors). Now look at the ways that we can choose colors where no two same-colored stripes share an edge. Begin by choosing any of the 3 colors for the vertical stripe. Once we choose a color for this stripe, none of the other stripes can be the same color. We can pick 1 of the 2 remaining colors for the middle horizontal stripe. The remaining stripes must both be the remaining color. This gives us only $3! = 6$ ways that no two same-colored stripes can share an edge. $6/81 = 2/27$.

3.330 There are $9C3 = 84$ combinations of 3 numbers. Only two common factors greater than 1 are possible: 2 and 3. There are 4 even numbers, and $4C3 = 4$ possible sets of 3 even numbers. There is one set of three numbers which have 3 as a common factor $\{3,6,9\}$. This means that there are 5 out of 84 combinations of 3 numbers which share a common factor greater than 1: $5/84$ is the probability of selecting one of these sets.

3.330 Three numbers are selected at random and without replacement from a bag containing each of the numbers 1 through 9. What is the probability that all three share a common factor greater than 1?

Example: Five boys and seven girls, are randomly assigned to three groups of four students each. What is the probability that one of the three groups will be all girls?

Reasoning: There are $12C4 = 495$ ways to select 4 students from a group of 12. There are $7C4 = 35$ ways to select 4 of the 7 girls. The probability of a randomly formed group consisting of all girls is therefore $35/495 = 7/99$. Because there are 3 groups formed, the probability of any one of those groups consisting of all girls is:

$$3 \cdot \frac{7}{99} = \frac{21}{99} = \frac{7}{33}.$$

This problem could have also been solved using compound probability by finding the probability of one group having four girls and then multiplying by three:

$$\frac{7}{12} \cdot \frac{6}{11} \cdot \frac{5}{10} \cdot \frac{4}{9} \cdot 3 = \frac{7}{99} \cdot 3 = \frac{7}{33}.$$

Example: The 26 letters of the alphabet are placed on tiles and randomly separated into two equal piles. What is the probability that the word MATH can be found in one of the two piles?

Reasoning: First, we consider the number of 13-letter piles selected from 26 letters: ${}_{26}C_{13}=10,400,600$. Then we consider the number of those piles that contain the letters MATH. This leaves room in the group for 9 more letters selected from a set of 22, or ${}_{22}C_9 =$

497,420. Dividing and simplifying gives us $11/230$. There are two piles of letters, so the probability of MATH being in one of the two piles is twice that:

$$2 \cdot \frac{11}{230} = \frac{11}{115}.$$

Using compound probability is more difficult here and I would not recommend it. The math looks like this when we use compound probability:

$${}_{13}C_4 \cdot \frac{4}{26} \cdot \frac{3}{25} \cdot \frac{2}{24} \cdot \frac{1}{23} \cdot 2 = \frac{11}{115}.$$

- 3.140 Three students are randomly selected from a group of twelve that includes twins Billy and Adam. What is the probability that Adam is selected but Billy is not?
- 3.140 There are $12C3 = 220$ ways to select three students from a group of 12. If Adam is selected, there are 10 students left (excluding Billy) and we need to choose two to pair with Adam. $10C2 = 45$ pairs can be selected, meaning there are 45 groups of 3 students which include Adam but not Billy out of the 220 possible groups. $45/220 = 9/44$.

3.240 A youth basketball team consists of 12 players. If they are randomly divided into 4 groups of 3 players each for passing drills, what is the probability that the 2 girls on the team will be placed in the same group?

3.240 There are many ways to approach this problem: First, the probability of two girls being in any one group is:

$$3\left(\frac{2}{12} \cdot \frac{1}{11} \cdot \frac{10}{10}\right) = \frac{1}{22} \text{ (the three is there because the}$$

group may be selected GGB, GBG, or BGG). There are four groups, making the probability $2/11$. Alternatively, there are $12C3 = 220$ ways to form a group and 10 ways to form a group with the two girls (2 girls plus 1 of the 10 remaining boys) for a probability of $1/22$ that any one of the groups has both girls.

Once again, 4 groups quadruples this probability to give us $2/11$. Finally (and perhaps easiest), consider the probability that one of the girls has already been assigned to a group. 2 of the 11 available spaces are in her group, making the probability $2/11$ that the second girl is placed with the first.

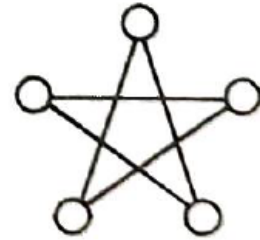
3.450 For any date in history, Luke can quickly tell you what day of the week it was. You choose three dates at random and Luke tells you the day of the week for each. What is the probability that at least two of the dates occurred on the same day of the week?

3.450 We will find the probability that no two dates fall on the same day of the week and subtract this from 1. The first date selected can be on any one of the seven days. The second date must fall on one of the six remaining days, and the third date must fall on one of the remaining 5 days for a probability of:

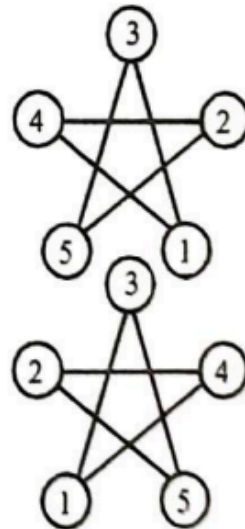
$\frac{7}{7} \cdot \frac{6}{7} \cdot \frac{5}{7} = \frac{30}{49}$, making the probability that at least two of the dates fall on the same day of the week

$$1 - \frac{30}{49} = \frac{19}{49}.$$

- 3.750 If the digits 1 through 5 are randomly arranged so that one digit fills each of the five circles, what is the probability that at least two consecutive integers will be connected by a line segment?



- 3.750 Start by placing the 3. Any arrangement can be rotated so that the three is located on top. When we place the three on top, this leaves 4 blanks to fill with 4 integers for $4! = 24$ possible ways to complete the diagram. To avoid consecutive integers, we can have a 1 and a 5 connected to the three as shown in two ways with the 2 and the 4 forced (depending on the placement of the 1 and the 5). This gives us just 2 ways to avoid consecutive integers (shown), so there are $22/24 = 11/12$ arrangements with at least one connected pair of consecutive integers.



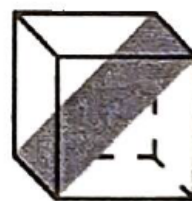
3.060 A bag contains 10 black blocks and 4 white blocks.
What is the probability of selecting 3 black blocks
from the bag without replacement?

3.060 Selecting blocks without replacement gives us:

$$\frac{10}{14} \cdot \frac{9}{13} \cdot \frac{8}{12} = \frac{30}{91}.$$

3.160 Four of the vertices of a cube are selected at random.
What is the probability that all four selected points
will be coplanar?

3.160 There are $8C4 = 70$ ways to select 4
of the 8 vertices of a cube. For all
four points to be on the same plane,
they can all be on the same face (6
ways), or on one of 6 diagonal planes slicing through
the cube (one is shown). $12/70 = 6/35$.



3.560 You are dealt three cards from a standard deck of 52 cards. What is the probability that exactly two cards are from the same suit?

3.560 In any suit, there are $13C2 = 78$ ways to select two cards, and there are 39 cards that can complete the set, making $78 \cdot 39 = 3,042$ ways to create a set with two cards from a given suit and one card from outside that suit. There are four suits, giving us $3,042 \cdot 4 = 12,168$ ways. There are $52C3 = 22,100$ ways to choose three cards from a set of 52. Simplifying $12,168/22,100$ give us **234/425**. Alternatively, consider the probability of choosing two cards of a given suit followed by one from a different suit:

$$\frac{13}{52} \cdot \frac{12}{51} \cdot \frac{39}{50} = \frac{39}{850}.$$

There are three ways to order

the selection of the suits (two hearts and a “not heart” can be selected HHX, HXH, or XHH). We must also multiply by the 4 choices for the suit that has two cards. This gives us:

$$3 \cdot 4 \cdot \frac{39}{850} = \frac{468}{850} = \frac{234}{425}.$$

With either method, this is an easy problem to mess up.

3.660 3 digits are selected from the set $\{2, 3, 4, 5, 6, 7, 8\}$ without replacement. What is the probability that the sum of the three digits is prime?

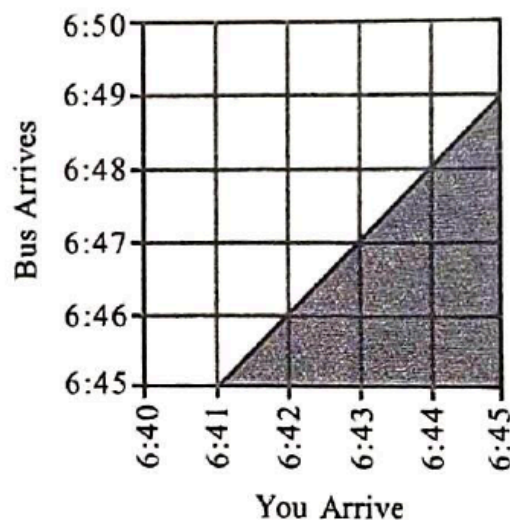
3.660 There are $7C3 = 35$ possible combinations. We are looking for combinations whose sum is prime. Be methodical. The smallest possible sum is 9 and the largest possible sum is 21. Primes between 9 and 21 are 11, 13, 17, and 19. Look for ways to get each:
 $11 = 2+4+5 = 2+3+6$ (2 ways),
 $13 = 2+5+6 = 2+4+7 = 2+3+8 = 3+4+6$ (4 ways),
 $17 = 2+7+8 = 3+6+8 = 4+6+7 = 4+5+8$ (4 ways),
 $19 = 4+7+8 = 5+6+8$ (2 ways).
This gives us a total of 12 out of 35 sums that are prime, or a probability of $12/35$.

3.760 Twenty-seven students each flip a fair coin. What is the probability that there are more heads showing than tails?

3.760 Consider the question, "What is the probability that there are more tails showing than heads?". The answer to both questions must be the same, and it is impossible to flip an equal number of heads and tails with 27 flips. The answer must therefore be $1/2$.

Example: The bus comes to a stop near your house every day at a random time between 6:45 and 6:50. You arrive at the bus stop at a random time between 6:40 and 6:45 every day and wait until the bus comes. What is the probability that you wait less than 4 minutes for the bus to arrive? Express your answer as a percent.

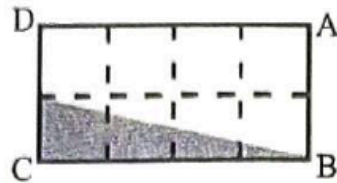
Reasoning: Set up a graph with your arrival time graphed against the arrival time of the bus. Shade the area where your wait time is less than 4 minutes.



The whole graph has an area of 25 units², and the shaded triangle is 8 units². Your probability of waiting less than 4 minutes is $\frac{8}{25}$ or 32%.

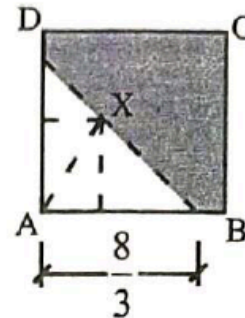
3.070 In rectangle ABCD, $AB = 1$ and $BC = 2$. Point X is selected at random within the rectangle. What is the probability that the area of triangle ABX is more than twice the area of triangle BCX?

3.070 If point X is selected within the shaded area, the height of triangle ABX will be greater than 4 times the height of triangle BCX, making its area greater than twice that of triangle BCX (its base is half that of triangle BCX). The shaded area is $\frac{1}{4}$ the total area.



3.370 Point X is selected at random within square ABCD of side length 3. What is the probability that quadrilateral ABXD has an area greater than 4 square units?

3.370 Consider the area of quadrilateral ABXD as the sum of the areas of triangles ADX and ABX. Each has a base length 3, so the sum of the heights of the triangles must be $\frac{8}{3}$ for the combined area to be greater than 4 (X must be anywhere within the shaded area). The area of the unshaded triangle is $(\frac{1}{2})(\frac{8}{3})^2 = \frac{32}{9}$ and the area of the entire figure is $\frac{81}{9}$, making the shaded area $\frac{49}{9}$. The probability that point X is within the shaded area is therefore

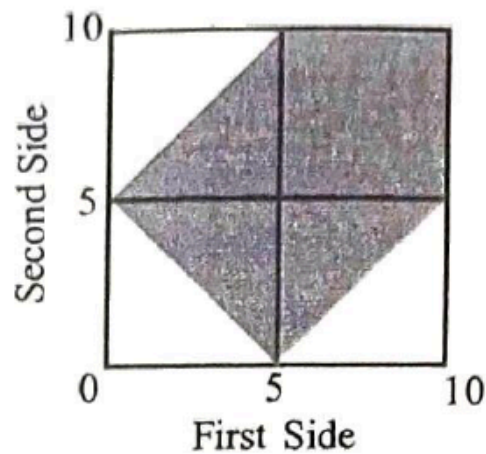


$$\frac{49/9}{81/9} = \frac{49}{81}.$$

3.470 One side of a triangle is 5cm long. Two (not necessarily integer) numbers are randomly selected between 0 and 10. What is the probability that the two numbers can be the other two sides of the triangle

3.470 The remaining sides of the triangle must have a sum that is greater than 5 and a difference that is less

than 5 as represented by the shaded area in the graph below. The shaded area is $\frac{5}{8}$ the total area.



3.090 The expiration date on a credit card is given as two digits for the month and two digits for the year. What is the expected value for the sum of the digits in the expiration date for a credit card that expires between 01/10 and 12/15 inclusive? Assume that each month and year occur with equal frequency.

3.090 We will do digit sums and expected value for the months and years separately to simplify things. The expected value for the digit sum of a month is $(1 + 2 + 3 + \dots + 9) + (1 + 0) + (1 + 1) + (1 + 2) = 51$. Because each month occurs the same number of times we just divide this by 12 to get 4.25. The expected value for the digit sum of a year between 10 and 15 is $[(1 + 0) + (1 + 1) + (1 + 2) + (1 + 3) + (1 + 4) + (1 + 5)]/6 = 21/6 = 3.5$. The expected sum of the digits is therefore $4.25 + 3.5 = 7.75$.

3.390 In five rolls of a standard die, what is the probability of rolling a “full house”, that is, two of one number and three of a different number?

3.390 There are $6^5 = 7,776$ ways to roll a set of 5 dice. We will use XXXOO to represent the full house, where X represents one number rolled and O represents a different number. There are $5C2 = 10$ arrangements of XXXOO, and we can choose from 6 numbers to use for X and 5 are left for O. This gives us $10 \cdot 6 \cdot 5 = 300$ of the 7,776 ways, or a probability of **25/648**.